

MATLAB CODE:

```
% Read an audio file
[x, fs] = audioread('file_example_WAV_2MG.wav');

% Set parameters
frame_size = 20; % Frame size in milliseconds
overlap = 15; % Overlap in milliseconds
threshold = 0.005; % ZCR threshold for classification

% Convert parameters to samples
frame_size_samples = round(frame_size * fs / 1000);
overlap_samples = round(overlap * fs / 1000);

% Calculate ZCR for each frame
zcr = zerocrossings(x, frame_size_samples,
overlap_samples);

% Classify frames into voiced, unvoiced, or silence based
on the threshold
voiced_frames = zcr > threshold;
unvoiced_frames = zcr <= threshold & zcr > 0;
silence_frames = zcr == 0;

% Display results
disp('Number of Voiced Frames:');
disp(sum(voiced_frames));

disp('Number of Unvoiced Frames:');
disp(sum(unvoiced_frames));

disp('Number of Silence Frames:');
disp(sum(silence_frames));

% Plot the audio signal with highlighted regions
t = (0:length(x)-1) / fs;

figure;

subplot(3,1,1);
plot(t, x);
```

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title('Audio Signal');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(3,1,2);
hold on;
plot(t(voiced_frames), zcr(voiced_frames), 'go',
'MarkerSize', 5);
plot(t(unvoiced_frames), zcr(unvoiced_frames), 'ro',
'MarkerSize', 5);
plot(t(silence_frames), zcr(silence_frames), 'bo',
'MarkerSize', 5);
hold off;
title('Zero Crossing Rate with Classification');
xlabel('Time (s)');
ylabel('ZCR');
legend('ZCR', 'Voiced', 'Unvoiced', 'Silence');

subplot(3,1,3);
plot(t, x);
hold on;
plot(t(voiced_frames), x(voiced_frames), 'go',
'MarkerSize', 5);
plot(t(unvoiced_frames), x(unvoiced_frames), 'ro',
'MarkerSize', 5);
plot(t(silence_frames), x(silence_frames), 'bo',
'MarkerSize', 5);
hold off;
title('Voiced, Unvoiced, and Silence Regions');
xlabel('Time (s)');
ylabel('Amplitude');
legend('Audio Signal', 'Voiced', 'Unvoiced', 'Silence');

% Separate the audio signal into voiced, unvoiced, and
silence regions
voiced_signal = x(voiced_frames);
unvoiced_signal = x(unvoiced_frames);
silence_signal = x(silence_frames);

```

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% Plot individual regions
figure;

subplot(3,1,1);
plot(t(voiced_frames), voiced_signal, 'go', 'MarkerSize',
5);
title('Voiced Region');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(3,1,2);
plot(t(unvoiced_frames), unvoiced_signal, 'ro',
'MarkerSize', 5);
title('Unvoiced Region');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(3,1,3);
plot(t(silence_frames), silence_signal, 'bo',
'MarkerSize', 5);
title('Silence Region');
xlabel('Time (s)');
ylabel('Amplitude');

% Function to calculate Zero Crossing Rate for each frame
function zcr = zerocrossings(x, frame_size, overlap)
num_frames = floor((length(x) - overlap) / (frame_size -
overlap));
zcr = zeros(1, num_frames);

for i = 1:num_frames
start_idx = (i - 1) * (frame_size - overlap) + 1;
end_idx = start_idx + frame_size - 1;
frame = x(start_idx:end_idx);

% Calculate Zero Crossing Rate
zcr(i) = sum(abs(diff(sign(frame)))) / (2 * frame_size);
end
end

```

OUTPUT:

AUDIOIE1CODE

Number of Voiced Frames:

2328

Number of Unvoiced Frames:

60

Number of Silence Frames:

0

FIGURE 1:

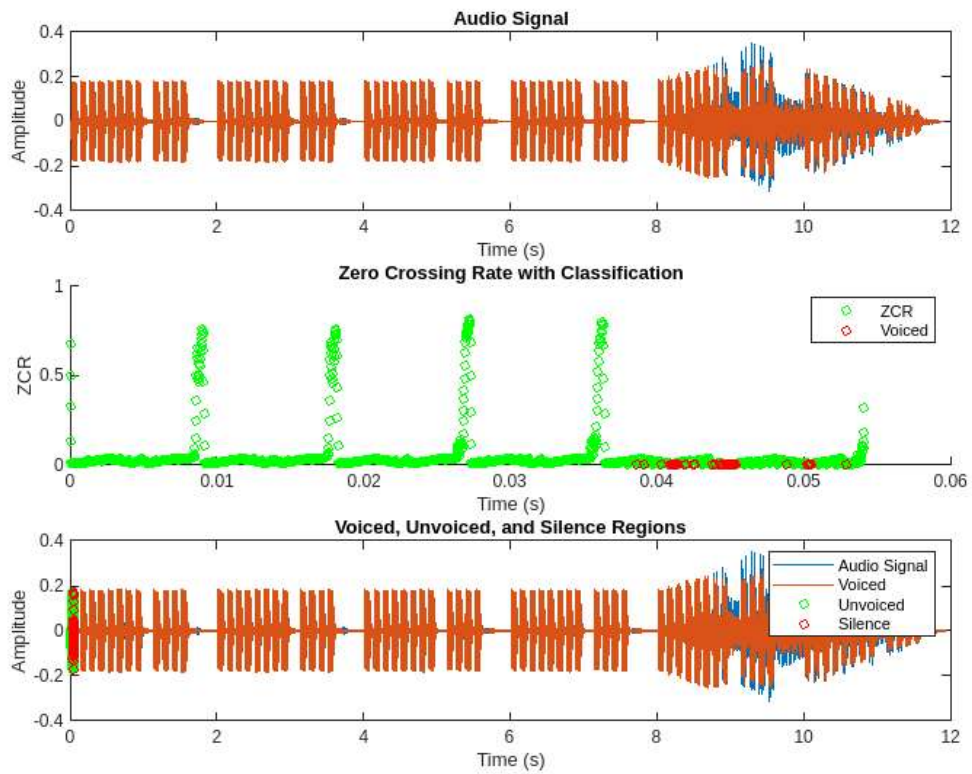
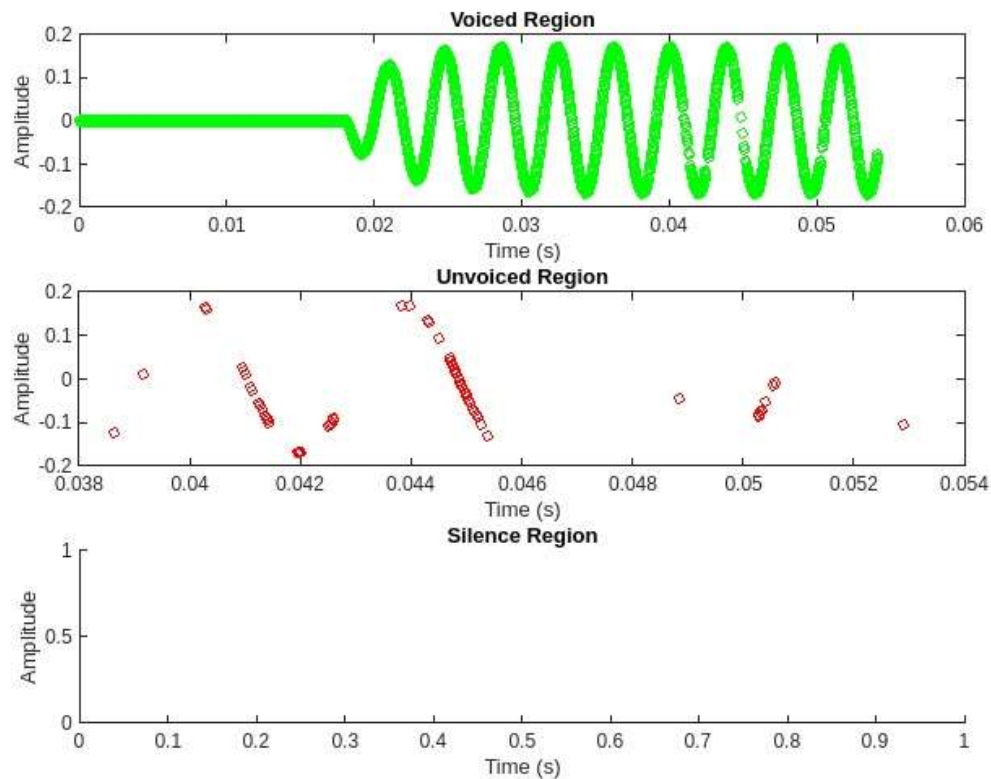


FIGURE 2:



MATLAB Code Summary:

This MATLAB code demonstrates the identification of voiced speech, unvoiced speech, and silence in an audio file using the Zero Crossing Rate (ZCR) technique.

Here's a brief summary:

1. Read Audio File:

Reads an audio file eg.('file_example_WAV_2MG.wav') and extracts the signal x and its sampling frequency fs .

2. Set Parameters:

- `frame_size`: Size of each frame in milliseconds.
- `overlap`: Overlap between frames in milliseconds.
- `threshold`: ZCR threshold for classification.

3. Calculate ZCR for Each Frame:

- The function zerocrossings is used to calculate the ZCR for each frame.
- ZCR is a measure of how many times the signal changes sign within each frame.

4. Classification:

- ❖ Frames are classified into three categories based on the ZCR and the defined threshold:
- Voiced frames: $ZCR > \text{threshold}$
- Unvoiced frames: $0 < ZCR \leq \text{threshold}$
- Silence frames: $ZCR = 0$

5. Plotting:

- Plots the audio signal with highlighted regions of voiced, unvoiced, and silence.
- Displays the ZCR with classification.
- Plots the separate regions of voiced, unvoiced, and silence signals.

Conclusion:

- The code effectively segments the audio signal into regions of voiced, unvoiced, and silence based on the Zero Crossing Rate.
- It provides visualizations to help understand where these segments occur within the audio.
- This experiment showcases a basic but fundamental technique in speech processing for segmenting different types of speech and silence.